

5. Conclusions

We summarise what this study establishes, the limitations under which it was established, and what remains to be done.

We have formulated the reconstruction of a charge stability diagram from sparse measurements as the probabilistic reconstruction of its transition lines, rather than of the sensor image. Given a small number of one-dimensional voltage sweeps, a U-Net returns, for every pixel of the gate-voltage plane, the probability that a transition line passes through it. The quantity that determines the charge occupation is therefore obtained in a single step, without an intermediate image or a separate line detector. Because the network is trained entirely on simulated devices, its training labels are exact and the amount of training data is not limited by experimental effort.

On 50 simulated test devices, the transition lines were recovered with $F1@1$ between 0.66 and 0.82 while sampling only 1.6–4.7 % of the voltage plane. The best budget, 8 rays of 60 points, reached $F1@1 = 0.82$ at a coverage of 4.7 %, and the score increased with the number of rays for nearly all budgets. The remaining errors are dominated by lines displaced by a single pixel rather than by missed lines, as shown by the rise of $F1$ from 0.47 at $\tau = 0$ to 0.82 at $\tau = 1$. To make these scores interpretable, the binarisation threshold was selected on validation data only, and every score is reported together with its tolerance and the fraction of the plane that this tolerance admits.

These results were obtained under idealised conditions. The devices were simulated with a noise-free constant-capacitance model, over a voltage window of fixed size, and training and test devices were drawn from the same capacitance intervals. The present work therefore establishes that the geometry of a double-dot stability diagram can in principle be recovered from rays covering a few percent of the plane; it does not yet demonstrate this on measured devices. In addition, the network outcome depends on training, as shown by the four budgets that had to be retrained, so small differences between individual budgets should not be over-interpreted.

Several directions follow from this work. Training on simulations that include noise and tunnel coupling, and testing on measured diagrams, is the essential next step. Beyond that, the method could be focused on the interdot transitions and triple points that tuning actually requires, informed by previously measured diagrams of the same device, and combined with fast RF readout, for which a small set of continuous voltage sweeps is naturally suited. Together, these steps would turn sparse ray measurements into a practical tool for the rapid

characterisation and re-tuning of quantum-dot devices.